ESTIMATION OF LAND SURFACE TEMPERATURE OVER A DECADE

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Land Surface Temperature (LST) Analysis of Delhi

This document presents a comprehensive analysis of Land Surface Temperature (LST) in Delhi, focusing on the peak summer months of May to June. Utilizing satellite imagery from Landsat 8, the study aims to understand temperature variations over the years 2015 and 2025. The findings highlight the impact of urbanization on temperature changes and the relationship between land use/land cover (LULC) and LST.

Introduction

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Land Surface Temperature (LST) is a critical parameter for assessing urban heat, climate change, and environmental conditions. It is derived from satellite data and provides insights into the thermal characteristics of the Earth's surface. This project calculates LST using satellite imagery to analyze temperature variations in Delhi, particularly during the hottest months of the year.

Study Area

The selected study area is **Delhi**, India. The analysis focuses on the months of **May to June** for the following reasons:

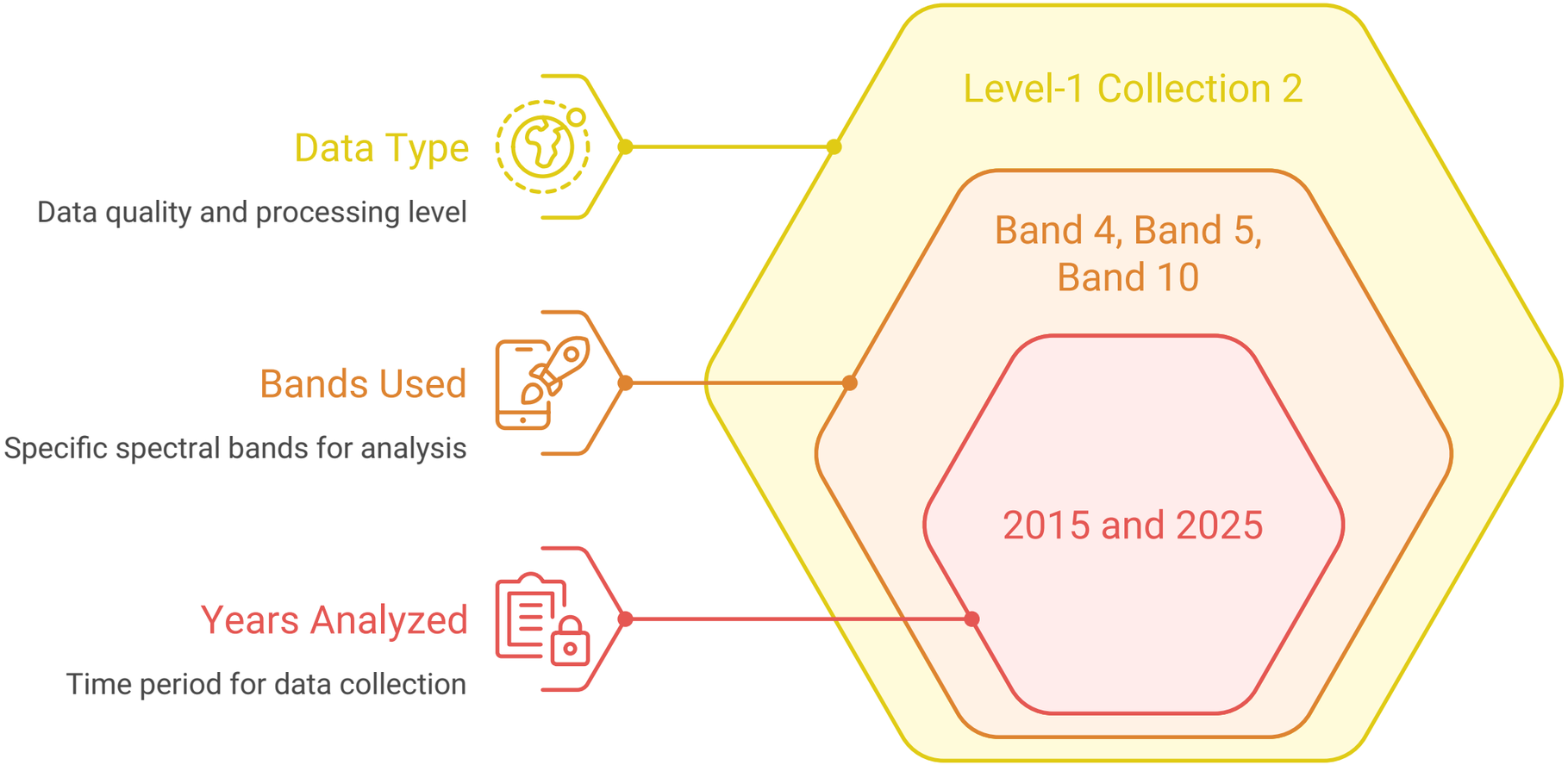
- These months represent peak summer, characterized by the highest temperatures.
- Understanding temperature variations during this period is crucial for assessing heat stress and the urban heat island effect.

Data Used

The analysis utilizes data from **Landsat 8**, specifically:

- **Data Type:** Level-1 Collection 2
- **Bands Used:**
    - Band 4 (Red)
    - Band 5 (NIR)
    - Band 10 (Thermal)
- **Additional File:** MTL.txt
- **Years Analyzed:** 2015 and 2025

Landsat 8 Data Used for Delhi LST Analysis



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Software & Methodology

Software Used: QGIS

The following steps were performed to calculate LST:

1. **TOA Radiance Calculation:** Conversion of raw satellite data to Top of Atmosphere (TOA) radiance.
2. **Brightness Temperature (BT) Calculation:** Deriving brightness temperature from thermal band data.
3. **NDVI Calculation:** Calculating the Normalized Difference Vegetation Index to assess vegetation cover.
4. **Proportion of Vegetation (PV):** Estimating the proportion of vegetation in the study area.
5. **Land Surface Temperature (LST) Calculation:** Using the brightness temperature and NDVI to compute LST.
6. **Zonal Statistics Analysis:** Analyzing LST across different land use categories.

Software & Methodology

1 TOA Radiance Calculation

- Data Type: Level-1 Collection 2
- Bands Used: Band 4 (Red), Band 5 (NIR), Band 10 (Thermal)
- Additional File: MTL.txt
- Years Analyzed: 2015 and 2025

2 Brightness Temperature (BT) Calculation

3 NDVI Calculation

4 Proportion of Vegetation (PV)

5 Land Surface Temperature (LST) Calculation

6 Zonal Statistics Analysis

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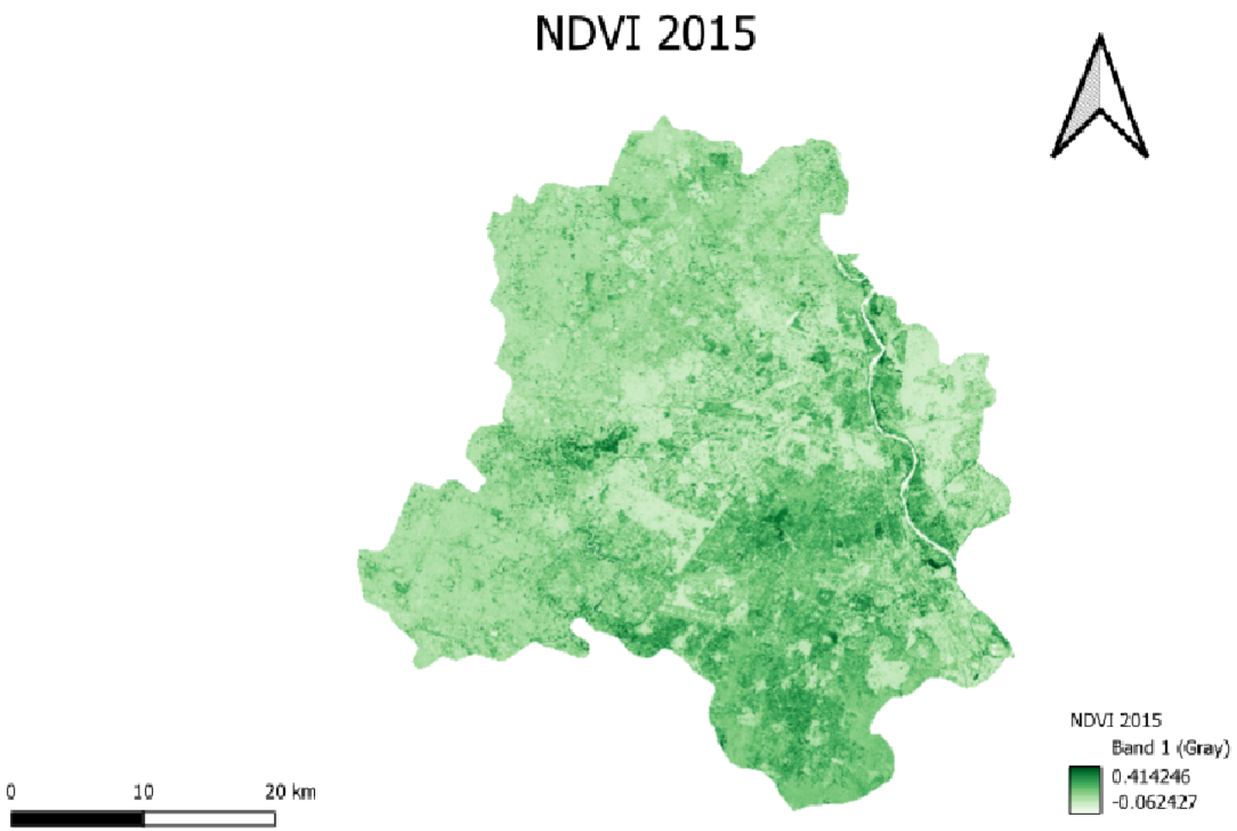
NDVI 2015



NDVI 2025

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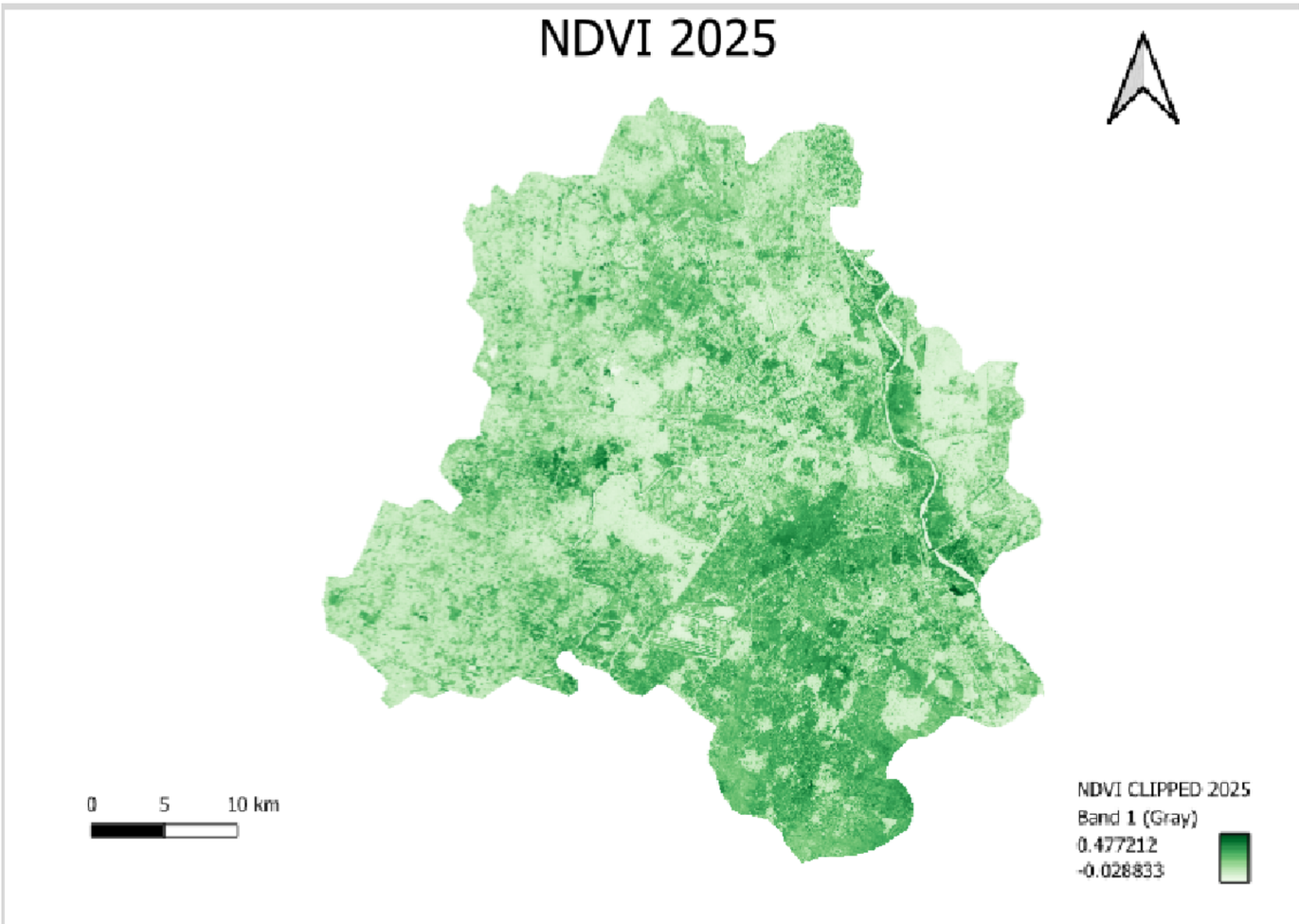

LST Map (2015)

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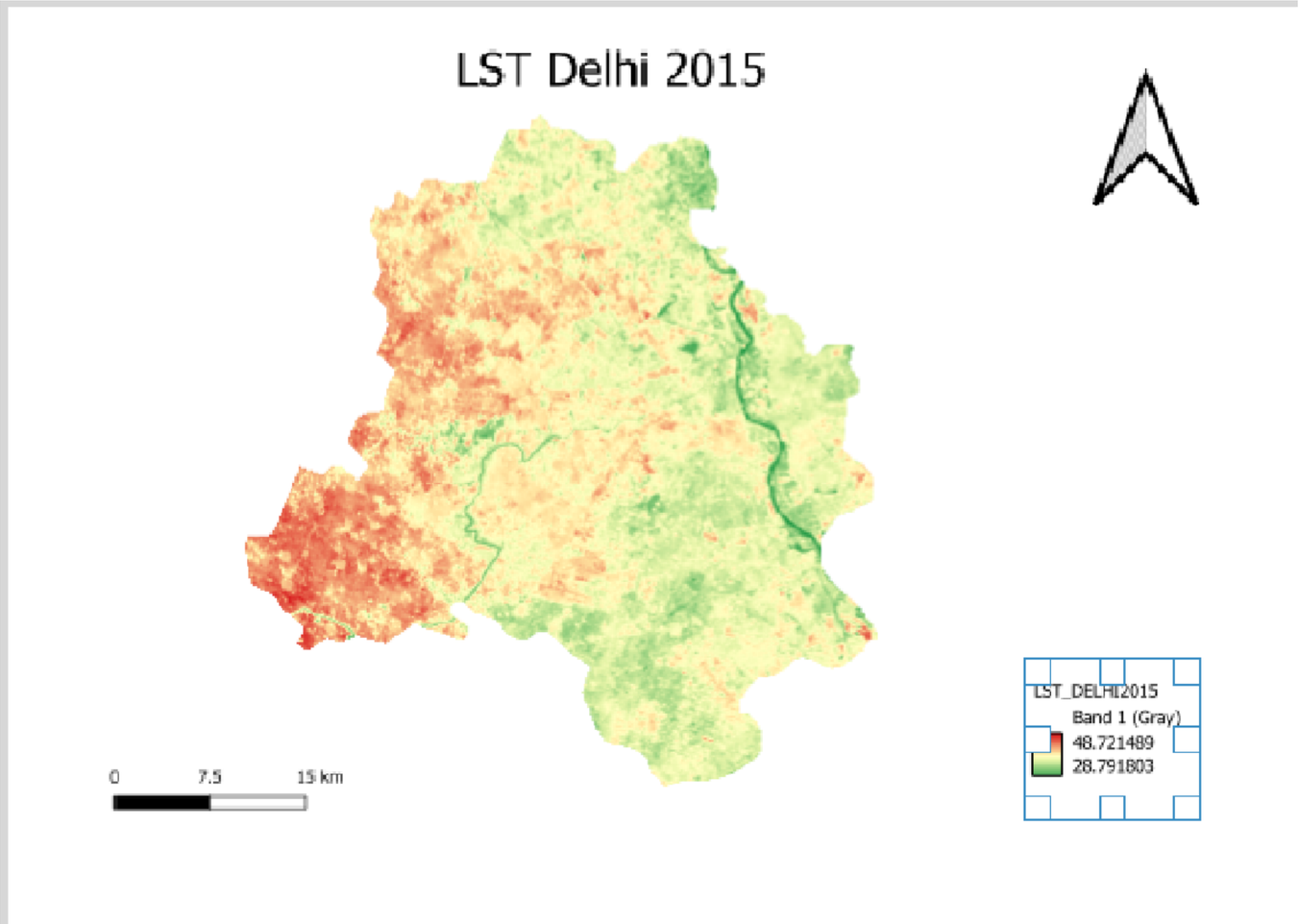

LST Map (2025)

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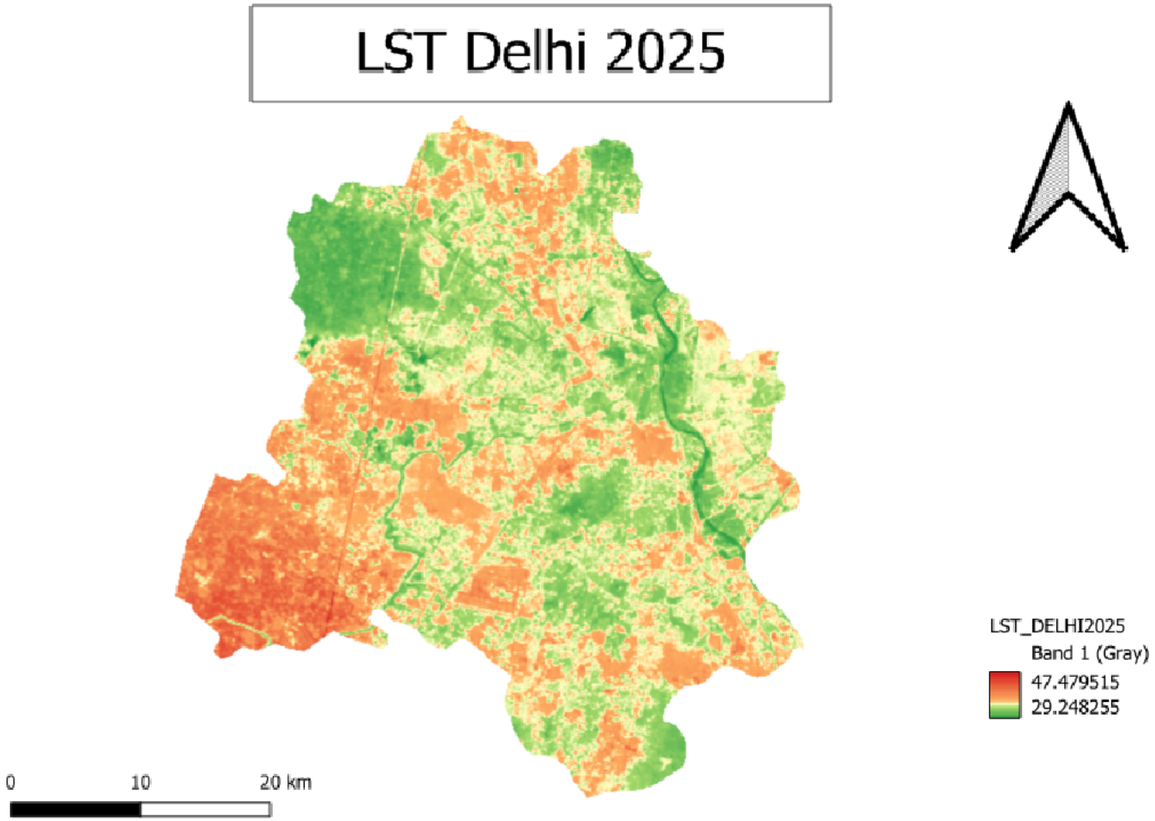

Comparison Table (2015 vs 2025)

| Parameter | 2015 | 2025 |

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Minimum Temperature	28.7918034	29.2482548
Maximum Temperature	48.721489	47.4795151

LULC and LST (2025)

	fid	Name	LST_mean
1	5	AGRICULTURE	35.1773795343...
2	4	BARREN LAND	35.6304637390...
3	2	BUILT UP	37.4459394257...
4	3	FOREST	33.8543380737...
5	1	WATER BODIES	30.1835769834...

Key Conclusions

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Parameter / Factor	Observation
LST Trend	Land Surface Temperature has shown a significant increase from 2015 to 2025.
Built-up Areas	Built-up areas exhibit the highest temperatures among all land use classes.
Vegetation Areas	Vegetated regions show relatively lower temperatures.
Water Bodies	Water bodies record the lowest surface temperatures.
Urbanization Impact	Urbanization is identified as a major factor contributing to the increase in temperature.
LULC Relationship	A direct correlation is observed between land use/land cover (LULC) and LST.
Minimum Temperature	Minimum temperature has increased in 2025 compared to 2015.

Thank you!